

94

1. (8 pts) Given the below integer values, write the corresponding **2-byte** binary and hex values:

	2-byte Binary value	2-byte Hex values
47	<u>0000 0000 0010 1111</u>	<u>002F</u> ✓
-13	<u>1111 1111 1111 0011</u>	<u>FFF3</u> ✓

2. (6 pts) Given the single byte value in hex as **D5** write the equivalent value if the single byte is interpreted as

a) an unsigned decimal value	<u>213</u> ✓	$16 \times 13 + 5 = 160 + 48 + 5 = 213$
b) a signed decimal value	<u>-43</u> ✓	$D5 \rightarrow 2A + 1 = 2B = 32 + 11 = 43$
c) as a single EBCDIC character	<u>N</u> ✓	$C1=A, D1=N$

3) (6 pts) Given the single byte in hex as **46** write the equivalent value if the single byte is interpreted as

a) an unsigned decimal value	<u>70</u> ✓	$16 \times 4 + 6 = 70$
b) a signed decimal value	<u>70</u> ✓	
c) a single ASCII character	<u>F</u> ✓	$A=41$

4. a) (8 pts) What are the eight 8-bit "workhorse" registers? AH, AL, BH, BL, CH, CL, DH, DL ✓
 b) (4) What are the four 16-bit "workhorse" registers? AX, BX, CX, DX ✓
 c) (4) What are the four 32-bit "workhorse" registers? EAX, EBX, ECX, EDX ✓
 d) (4) What are the four 64-bit "workhorse" registers? RAX, RBX, RCX, RDX ✓

5. a) (2) What are the two 16-bit index registers? SI, DI ✓
 b) (2) What are the two 32-bit index registers? ESI, EDI ✓

6. a) (3) What are the three 16-bit pointer registers? BP, SP, IP ✓
 b) (3) What are the three 32-bit pointer registers? EBP, ESP, EIP ✓

7. (8) How many bytes get assigned when using the below directives?

BYTE	<u>1</u>	DWORD	<u>4</u>
QWORD	<u>8</u>	WORD	<u>2</u>

8. (3) There are 3 different "areas" of storage associated with an assembly language program. What are

these three areas? ~~stack~~ memory, ~~data~~ registers, and ~~code~~ disk -3

9. (3) What are the 3 directives which inform the assembler that the statements following the directive belongs to that particular segment?

.stack, .data, .code

10. (18) Give the actual contents of memory when the data gets loaded into memory. Keep in mind that ALL information that is non-character is stored in reverse-byte order. Complete the values for the location counter:

Loc. Ctr Identifier

<u>0004</u>	A	WORD 2	27,15,8	<u>1B 00 0F 00 08 00</u> ✓
<u>000A</u>	B	WORD 2	3 dup (-1)	<u>FFFF FFFF FFFF</u> ✓
<u>0010</u>	C	DWORD 4	19,35	<u>13 00 00 00 23 00 00 00</u> ✓
<u>0018</u>	E	BYTE 1	5,7,'AB'	<u>05 07 41 42</u> ✓
<u>001C</u>	D	QWORD 8	25	<u>19 00 00 00 00 00 00 00</u> ✓
<u>0024</u>	E	BYTE 1	3 dup (-29) ^{ID} _{E2 → E3}	<u>E3 E3 E3</u> ✓
<u>0027</u>		stName BYTE 1	2 dup ('ABC')	<u>41 42 43 41 42 43</u> ✓

11. (8 pts) How many bytes are associated with each of the below terms (you can use a power of 2 in your answers)

- | | | | | | |
|--------------|------------------------------|--------------|------------------------------|-------------|------------------------------|
| a) kilobyte | <u>2^{10}</u> ✓ | d) yottabyte | <u>2^{80}</u> ✓ | g) petabyte | <u>2^{50}</u> ✓ |
| b) megabyte | <u>2^{20}</u> ✓ | e) gigabyte | <u>2^{30}</u> ✓ | h) terabyte | <u>2^{40}</u> ✓ |
| c) zettabyte | <u>2^{70}</u> ✓ | f) Exabyte | <u>2^{60}</u> ✓ | | |

yes!!!

(Continue on the back)

12. (6 pts) When your program gets loaded into memory certain registers get loaded with the contents of the different "pieces" of your program. Suppose that the below memory dump shows the contents of the registers just before the very first instruction is executed. Answer the following questions:

- a) What is the address of the very first instruction to be executed? 00401058 ✓
- b) What is address of the very first byte beyond the top of the stack? 00401057 ✓
- c) Given that the stack is 100h, what is the address of the very first byte in the stack? (NOT the top of the stack)

00400F57 ✓

Registers	
EAX	76d33388
EBX	7efde000
ECX	00000000
EDX	00401058
ESI	00000000
EDI	00000000
EIP	00401058
ESP	0018ff8c
EBP	0018ff94

13. (4 pts) Given the attached disk layout, assume that the os supervisor is focused on the C: drive in the TD "room" (folder).

- a) What prompt is displayed on the screen?

C:\Temp\FIN\TD\ ✓

- b) What is the path (thru the "doors") to the INC folder?

..\..\..\usr\INC ✓